

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES
Final Exam

M.CA. (Winter Semester-2023)

Shoreya Pal

Course Name: Computer Architecture

Maximum Marks: 60

Course Code: CS 106

Total Time: 3.0 hours

Q1. (a) What is the difference between X86 and ARM type of Computer Architecture?

(b) Explain the concept of parallelism in Computer System.

(c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.2 and ii) reduce CPI instruction type B to 1.5. Which one is better?

Type	CPI	Frequency
A	1.5	50%
B	2.5	30%
C	2.1	20%

(3+2+5)

Q2. (a) What is flash memory and which part of memory hierarchy it is placed? Explain the term locality of reference.

(b) What conclusion can one draw from the following observations in terms of computing performance

Type	CPU A (in Sec.)	CPU B (in Sec.)	CPU C (in Sec.)
Program A	60	40	30
Program B	510	220	90
Total Time	570	260	120

(4+6)

Q3. (a) What is a Vector Processor?

(b) The great advantage of pipelining is the performance improvement with a shorter clock time.

Element	Register Clk to q	Register Set Up	Mux.	ALU	Mem. Read	Mem. Write	Register File Read	Register File Set Up
Parameter	t clk-to-q	t Setup	t Mux.	t ALU	t MEM read	t MEM write	t RF Read	t RF setup
Delay (ps)	30	20	25	200	250	200	150	5

i) What was the clock time and frequency of a single cycle CPU?

ii) What is the clock time and frequency of a pipelined CPU?

iii) What is the speed-up?

(4+6)

Q4. (a) What are the guiding parameters for defining the memory Hierarchy?

(b) Consider two different types of memory hierarchy. In the first case, consider a three level memory hierarchy with separate instruction and data cache in level 1 while the main memory is at level 2. In the second case, consider a three level memory hierarchy with separate instruction and data cache in level 1 and level 2 respectively. The main memory is at level 3.

Calculate the average access time in both types of memory hierarchy for the following parameters:

- i) Clock cycle is 2 ns
- ii) Miss cycle is 10 clock cycle
- iii) 2 percent of instruction is not found in Instruction cache
- iv) 9 percent of instruction is not found in Data cache
- v) 25 percent of the total memory access for the data
- vi) Cache Access Time (including hit detection) is 1 clock cycle.

Which one of the memory configuration is better?

(2+8)

Q5. (a) What is the difference between Deep and Shallow pipelining technique?

(b) For the following reservation table

- (i) What are the forbidden latencies
- (ii) Show the state transition diagram
- (iii) List all the simple and greedy cycles
- (iv) Determine the optimum MAL
- (v) Determine the pipeline throughput with $\tau=20\text{ns}$

	1	2	3	4	5	6
S1	X		X	X	X	
S2		X		X		X
S3		X	X		X	

	1	2	3	4	5
S1	Y		Y	Y	Y
S2		Y		Y	
S3	Y	Y			

(2+8)

Q6. (a) What are the different ways for cache optimization techniques.

(b) Suppose that in 1000 memory references there are 60 misses in L-1 cache are 30 misses in L-2 cache and cache are 10 misses in L-3 cache. What are various miss rates?

Assume that Miss Penalty $L-2$ is 180 clock cycles, Hit time $L-1$ is 1 clock cycle, Hit time $L-2$ is 6 clock cycle, Miss Penalty $L-3$ is 90 clock cycles, Hit time $L-3$ is 12 clock cycle

What will be the average memory access Time?. Ignore impact of write.

(4+6)